Breathing and exchange of gases

Introduction and Respiratory Organs

1. Given below are two statements:

Statement I: Mechanisms of breathing vary among different groups of animals depending mainly on their habitats and level of organisation.

abitats and level of organisation. Statement II: The process of exchange of O_2 from the atmosphere with CO_2 produced by the cells is called breathing.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both statement I and II are incorrect.
2) Statement I is correct, but statement II is incorrect.
3) Statement I is incorrect, and statement II is correct.
4) Both statement I and II are correct.

2. What is the reason behind calling respiration direct in insects?

1) The tissues exchange O_2/CO_2 directly with the air outside through body surface
2) Tracheal tubes exchange O_2/CO_2 directly with the haemocoel which then exchange with tissues
3) The cell exchange O_2/CO_2 directly with the air in the tubes
4) The tissues exchange O_2/CO_2 directly with coelomic fluid

3. Which of the following is the correct pathway of air in the respiratory system?

1) External nostril → nasal chamber → nasal passage →larynx pharynx → trachea → primary bronchi →secondary bronchi → tertiary bronchi → initial bronchioles → terminal bronchioles → duct of alveoli → alveoli
2) External nostril →nasal passage → nasal chamber→ pharynx → larynx → trachea → initial bronchioles →terminal bronchioles → primary bronchi → secondary bronchi → tertiary bronchi → duct of alveoli → alveoli
3) External nostril → nasal chamber → nasal passage→ larynx → pharynx → trachea → initial bronchioles → terminal bronchioles → primary bronchi → secondary bronchi → tertiary bronchi→ duct of alveoli→ alveoli
4) External nostril → nasal passage → nasal chamber →pharynx → larynx → trachea → primary bronchi → secondary bronchi → tertiary bronchi →→ initial bronchioles → terminal bronchioles → duct of alveoli alveoli

4. The lower invertebrates like sponges perform (i). Among vertebrates, fish use (ii), whereas reptiles, birds and mammals use (iii).

Fill in the blanks with the correct option:

	(i)	(ii)	(iii)
1)	Osmosis	Book lungs	Lungs
2)	Simple diffusion	Gills	Lungs
3)	Simple diffusion	Lungs	Gills
4)	Osmosis	Lungs	Gills

5. Which of the following is correct w.r.t. trachea?

1) Trachea is a straight tube.
2) Trachea divides into a pair of primary bronchi at the level of 5th thoracic vertebra.
3) Cartilaginous rings in trachea are incomplete.
4) All of the above

6. Which of the following part of the respiratory tract does not have cartilaginous rings in its wall?

1) Trachea
2) Primary bronchi
3) Secondary bronchi
4) Alveoli

7. Match the following columns and choose the correct option.

Column I		Column II	
1	Coelenterates	a	Lungs
2	Branchial respiration	b	Entire body surface
3	Insects	c	Gills
4	Pulmonary respiration	d	Tracheal tubes

1) 1-c, 2-d, 3-b, 4-a
2) 1-b, 2-c, 3-d, 4-a
3) 1-b, 2-d, 3-a, 4-c
4) 1-b, 2-c, 3-a, 4-d

8. Read the following statements and choose the correct option.

i. Gills are used for the exchange of gases in aquatic arthropods.
ii. Amphibians like frogs can respire through their moist skin.
iii. Epiglottis prevents the entry of food into the larynx.
iv. Nasal chamber opens into pharynx.

How many statements are correct?

1) One
2) Three
3) Two
4) Four

9. All of the following form conducting part of the respiratory tract, except

1) Trachea
2) Secondary bronchi
3) Alveoli
4) Terminal bronchiole

10. Read the following statements. Which of the following are correct?

(i) Lungs are covered with double-layered membrane called pleura.
(ii) Cardiac notch is present in right lung.
(iii) The alveolar ducts and alveoli constitute the exchange part of the respiratory tract.
(iv) Left lung has three lobes, while the right lung has two.

1) (i) and (ii)
2) (i) and (iii)
3) (ii) and (iv)
4) (iii) and (iv)

11. _____ is a flap of cartilage which is depressed during swallowing to cover the opening of larynx.

1) Glottig
2) Epiglottis
3) Cricoid
4) Trachea

12. Which of the following is the correct sequence of respiration in our body?

(i) Exchange of gases between deoxygenated blood and alveoli.
(ii) Utilization of oxygen by the cells.
(iii) Exchange of gases between oxygenated blood and tissue.
(iv) Inhalation of air.

1) (i), (iii), (ii), (iv)
2) (iv), (i), (iii), (ii)
3) (iv), (iii), (ii), (i)
4) (iv), (ii), (iii), (i)

13. Which. of the following is incorrect w.r.t. human respiratory system?

1) Larynx is the sound box in humans.
2) The nasal chambers open into larynx, the common passage for food and air.
3) We have a pair of external nostrils opening out above the upper lips.
4) Inner surface of bronchi and bronchioles is lined by ciliated epithelium.

14. Which of the following statements is incorrect regarding respiratory system?

1) Each terminal bronchiole give rise to a network of bronchi.
2) The alveoli are highly vascularised.
3) The lungs are covered by a double-layered membrane.
4) The pleural fluid reduces friction on the lung surface.

15. Which of the following is the function of pleural fluid?

1) Helps in exchange of gases
2) Expands the volume of thoracic cavity
3) Reduces friction on the lung-surface
4) Provide protection to the alveoli

16. Which of the following are functions of conducting part of the lungs.

i. Transport of atmospheric air to the alveoli.
ii. Humidifies and brings the air to body temperature.

iii. Clears atmospheric air from foreign particles.
iv. Exchange of gases between alveoli and blood.
1) i and ii
2) ii and iii
3) iii and iv
4) i, ii and iii

Mechanism of Breathing

17. Which of the following muscles contract during inspiration?
1) Diaphragm
2) Internal intercostal muscles
3) External intercostal muscles
4) Both (1) and (3)

18. The volume of air inspired or expired during normal respiration is called
1) Inspiratory capacity
2) Tidal volume
3) Inspiratory reserve volume
4) Vital capacity

19. A spirometer can be used to directly measure
1) Residual volume
2) Tidal volume
3) Functional residual capacity
4) All of the above

20. Which of the following is an incorrect statement?
1) Breathing involves two stages: inspiration during which atmospheric air is drawn in and expiration by which alveolar air is released out.
2) We have the ability to increase the strength of inspiration and expiration with the help of additional muscles in the abdomen.
3) Inspiratory capacity refers to the total volume of air a person can expire after a normal inspiration.
4) Relaxation of the diaphragm and the inter-costal muscles returns the diaphragm and sternum to their normal positions and reduce the thoracic volume and thereby the pulmonary volume.

21. Match the following columns and choose the correct option.

Column I		Column II	
1	Tidal Volume	a	RV + ERV
2	Expiratory capacity	b	ERV + TV
3	Functional residual capacity	c	IC-IRV
4	Expiratory reserve volume	d	VC-[TV+ IRV]

1) 1-c, 2-b, 3-a, 4-d
2) 1-d, 2-b, 3-a, 4- c
3) 1-b,2-a, 3-d, 4-c

4) 1-d, 2 – c, 3 – a, 4 -b

22. _____ is the volume of air accommodated in our lungs at the end of forced inspiration.

1) Residual volume
2) Vital capacity
3) Total lung capacity
4) Functional residual capacity

23. Which among the mentioned muscles are involved in the normal breathing in humans?

1) External intercostals muscles only
2) Diaphragm and intercostals muscles
3) External and internal intercostals muscles
4) Diaphragm and abdominal muscles

24. The lowest mechanical efforts are required in the respiration during

1) Vital capacity
2) Expiratory resetve volume
3) Inspiratory reserve volume
4) Tidal volume

25. Choose the correct statement with respect to normal breathing.

1) Expiration is a passive process while inspiration is an active process.
2) Inspiration and expiration are both active processes.
3) Expiration is an active while inspiration is a passive process.
4) Inspiration and expiration are both passive processes.

26. Which of the following event is observed during normal expiration?

1) Contraction of diaphragm
2) Contraction of external intercostal muscles
3) Relaxation of diaphragm
4) Rise in the volume of thoracic cavity

27. Identify the wrong statement:

1) Breathing rate is higher in children than in adults.
2) Expiratory capacity is the sum total of tidal volume and expiratory reserve volume.
3) IC= TV + IRV
4) The average residual volume is 2000-2500 mL.

28. Which of the following is incorrect w.r.t. total lung capacity (TLC)?

1) IRV + TV + ERV + RV
2) IC + FRC
3) EC+ FRC
4) VC+RV

Exchange of Gases and Transport of Gases

29. The primary site (s) of exchange of gases is/are
1) Trachea
2) Alveoli
3) Nasal chamber
4) Bronchioles

30. Which of the following factor(s) can affect the rate of diffusion?
i. Pressure gradient
ii. Concentration gradient
iii. Solubility of the gases
iv. Thickness of the membranes
1) i and ii
2) iii and iv
3) i, ii and iii
4) i, ii, iii and iv

31. Identify the wrong statement:
1) In lungs, gaseous exchange takes place by diffusion.
2) The solubility of oxygen is 20–25 times higher than that of carbon dioxide.
3) The thickness of diffusion membrane is less than a millimetre.
4) The amount of CO_2 that can diffuse through the diffusion membrane per unit difference in partial pressure is much higher compared to that of O_2.

32. Read the following statements:
(i) Both the lining of alveoli and capillaries have squamous epithelium.
(ii) A molecule of haemoglobin carries four oxygen molecules.
(iii) The globin component of haemoglobin has iron to carry oxygen.
(iv) The partial pressure of oxygen in the inhaled air is 104 mmHg.

Which of the above statements are incorrect?
1) (i) and (ii)
2) (iii) and (iv)
3) (i) and (iii)
4) (ii) and (iv)

33. The diffusion membrane is made up of
1) Thin squamous epithelium of alveoli
2) Endothelium of alveolar capillaries.
3) Basement substance
4) All of the above

34. The oxyhaemoglobin dissociation curve shifts to left, if
1) pH falls
2) pH increases
3) Levels of CO_2 increases
4) Temperature increases

35. Which of the following is observed if the blood becomes acidic?

1) Oxygen-carrying capacity of blood increases.
2) RBC count increases.
3) Oxygen-carrying capacity of blood decreases.
4) Less oxygen is released in tissues.

36. Carbon monoxide combines with haemoglobin to form

1) Carbaminohaemoglobin
2) Carboxyhaemoglobin
3) Oxyhaemoglobin
4) Deoxyhaemoglobin

37. Match the following columns and choose the correct option.

Column I		Column II	
1	Systemic veins	a	Oxygenated blood
2	Systemic arteries	b	Deoxygenated blood
3	Pulmonary artery		
4	Pulmonary vein		

1) 1-a, 2-b, 3-b, 4-a
2) 1-b, 2-a, 3-b, 4-a
3) 1-b, 2- a, 3- a, 4-b
4) 1-a, 2- b, 3- a, 4-b

38. Which of the following option is correct wr.t. the transport of CO_2 with haemoglobin?

1) CO_2 is carried by haemoglobin as carbaminohaemoglobin.
2) 20-25% of CO_2 is transported with haemoglobin.
3) CO_2 combines reversibly with haemoglobin.
4) All statements are correct.

39. hen pCO_2 is high and pO_2 is low as in the tissues _____ binding of carbon dioxide occurs with haemoglobin. Fill in the blank.

1) Less
2) More
3) No
4) Very little

40. Which of the following is an incorrect statement w.r.t. CO_2 transport?

1) Every 100 mL of deoxygenated blood delivers approximately 4 mL of CO_2 to the alveoli.
2) RBCs contain very high concentration of the enzyme, carbonic anhydrase.
3) Carbonic anhydrase is absent in blood plasma.
4) All of the above.

41. Read the following statements and identify the wrong one w.r.t. transport of gases.

1) Pressure contributed by an individual gas in a mixture of gases is called partial pressure.
2) Haemoglobin has the ability to bind reversibly with oxygen.

3) Increase in CO_2 concentration increases the oxygen affinity of haemoglobin in tissues.
4) pCO_2 of alveolar air is 40 mmHg.

42. The partial pressure of oxygen (pO_2) of deoxygenated and oxygenated blood, respectively, are
1) 40 mmHg and 95 mmHg
2) 95 mmHg and 40 mmHg
3) 104 mmHg and 40 mmHg
4) 45 mmHg and 104 mmHg

43. The partial pressure of oxygen (pO_2) in deoxygenated blood is
1) More than the pO_2 in alveolar air
2) Equal to the pCO_2 in alveolar air
3) Less than pCO_2 in atmospheric air
4) More than pCO_2 in deoxygenated blood

44. Which of the following is an incorrect statement?
1) The pO_2 of alveolar air is less than the pO_2 of atmospheric air.
2) The pCO_2 in oxygenated blood is equal to the pCO_2 in alveoli.
3) The pCO_2 of atmospheric air is more than the pCO_2 of alveolar air.
4) The pCO_2 at tissue is equal to the pCO_2 in deoxygenated blood.

45. Which of the following is an incorrect statement?
1) A sigmoid curve is obtained when percentage saturation of haemoglobin with O_2 is plotted against pO_2
2) The oxygen dissociation curve is highly useful in studying the effect of factors like pCO_2, H^+ concentration, etc. on oxygen affinity of haemoglobin.
3) In the alveoli, where there is high pO_2 low pCO_2 lesser H^+ concentration and lower temperature, the factors are favourable for breakdown of oxyhaemoglobin.
4) O_2 gets bound to haemoglobin in the lung surface and gets dissociated at the tissues.

46. Identify the correct set of statements:
(a) The diffusion membrane is made up of two major layers.
(b) The total thickness of diffusion membrane is much less than a millimetre.
(c) At the alveolar site where pCO_2 is low, H_2CO_2 leads to the formation of CO_2 and H_2O.
(d) About 97 per cent of O_2 is transported by RBCs in the blood and the remaining 3 per cent of O_2 is carried in a dissolved state through the plasma.

Choose the correct answer from options given below:
1) a and b only
2) b, c and d only
3) a, b, d only
4) only d

Regulation of Respiration and Disorders of Respiratory System

47. Neural signals from pneumotaxic centre
1) Increases the duration of inspiration
2) Reduces the duration of inspiration
3) Alters the respiratory rate
4) Both (2) and (3)

48. Chemosensitive area situated adjacent to the respiratory rhythm centre is highly sensitive to
1) CO_2
2) Hydrogen ions
3) Both (1) and (2)
4) Oxygen only

49. Identify the wrong statement w.r.t. the regulation of respiration.
1) Pneumotaxic centre can moderate the functions of respiratory rhythm centre.
2) Receptors associated with aortic arch and carotid artery can recognize changes in oxygen concentration and send signals to the rhythm centre for remedial actions.
3) Human beings have a significant ability to maintain the respiratory rhythm.
4) Human beings have a significant ability to maintain the respiratory rhythm.

50. Chemoreceptors located in medulla are stimulatedby
1) Increased CO_2 in arterial blood
2) Decreased CO_2 in arterial blood
3) Increased O_2 in arterial blood
4) Decreased O_2 in arterial blood

51. In industries involving grinding or stone breaking, chronic exposure to dust particles causes
1) Damage to alveolar walls
2) Spasm of bronchial muscles
3) Accumulation of fluid in pleural cavity
4) Inflammation leading to fibrosis

52. The allergic disease in which patient experiences difficulty in breathing, causing wheezing due to inflammation of bronchi and bronchioles is
1) Emphysema
2) Common cold
3) Asthma
4) Diphtheria

53. Specialised centres in the brain regulate respiratory process. Upon stimulation, which among the following Clisted centre can reduce inspiration duration?
1) Chemosensitive centre
2) Apneustic centre

3) Medullary inspiratory centre
4) Pneumotaxic centre

54. Match the following columns and choose the correct option

Column I		Column II	
1	Occupational respiratory disorders	a	Proliferation of fibrous tissues
2	Asthma	b	Alveolar walls are damaged
3	Emphysema	c	Grinding or stonebreaking industries
4	Fibrosis	d	Inflammation of bronchi and bronchioles

1) 1-c, 2-b, 3-d, 4- a
2) 1-a, 2-b, 3- c, 4-d
3) 1- a, 2-b, 3- d, 4-c
4) 1-c, 2- d, 3- b, 4- a

55. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R).

Assertion (A): Pneumotaxic centre can moderate the functions of respiratory rhythm centre.

Reason (R): Neural signal from this centre can reduce the duration of inspiration.

In the light of the above statements, choose the correct answer from the options given below:

1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.
4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

56. Rammesh was a chain smoker and ultimately detected with emphysema (a respiratory disorder). In his case:

1) The plasma membrane must be damaged
2) The respiratory muscles must be damaged
3) The alveolar walls must be damaged
4) The bronchioles must be damaged

Answer Key

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
4	3	4	2	4	4	2	4	3	2	2	2	2	1	3
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
4	4	2	2	3	1	3	2	4	1	3	4	3	2	4
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
2	2	4	2	3	2	2	4	2	3	3	1	2	3	3
46	**47**	**48**	**49**	**50**	**51**	**52**	**53**	**54**	**55**	**56**				
2	4	3	2	1	4	3	4	4	4	3				